

EEE - Presentation

AC - DC Resistance

$$I_D = I_S \left(e^{\frac{V_D}{n V_T}} - 1 \right)$$

$$\frac{d}{dV_D} (I_D) = \frac{d}{dV_D} \left\{ I_S \left(e^{\frac{V_D}{n V_T}} - 1 \right) \right\}$$

$$\frac{1}{r_D} = I_S e^{\frac{V_D}{n V_T}} \cdot \frac{1}{n V_T}$$

$$\frac{1}{r_D} = \frac{I_S}{n V_T} \left(e^{\frac{V_D}{n V_T}} \right)$$

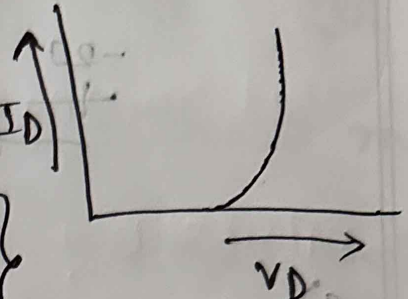
$$e^{\frac{V_D}{n V_T}} - 1 = \frac{I_D}{I_S}$$

$$e^{\frac{V_D}{n V_T}} = \frac{I_D}{I_S} + 1$$

$$= \frac{I_D + I_S}{I_S}$$

$$\frac{1}{r_D} = \frac{I_S}{n V_T} \times \frac{I_D + I_S}{I_S}$$

$$= \frac{1}{r_D} = \frac{1}{n V_T} \cdot I_D$$

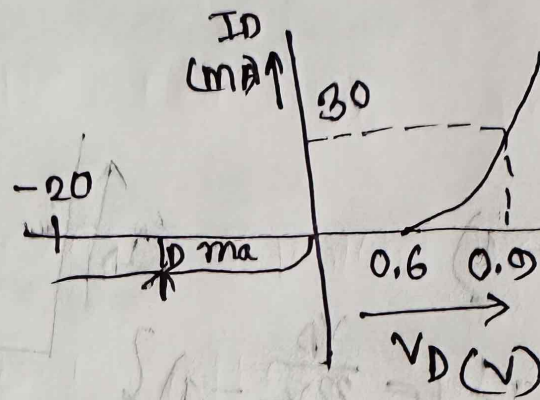


$$\begin{aligned} r_D &= \frac{n V_T}{I_D} \\ &= \frac{1 \times 0.026}{0.01} \\ &= \frac{0.026}{0.01} \end{aligned}$$

$$r_D = \frac{0.026}{I_D}$$

$$= \frac{0.026}{0.01} = 2.6 \Omega$$

Determine the DC resistance levels:



a) $I_D = 4 \text{ ma}$

b) $I_D = 30 \text{ ma}$

c) $V_D = -20 \text{ v}$

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(a) $\frac{0.6}{4 \times 10^{-3}} = 150 \Omega$

(b) $\frac{0.9}{30 \times 10^{-3}} = 30 \Omega$

(c) $\frac{20}{10 \times 10^{-6}} = 2 \text{ m} \Omega$